

Daily Geometry Drill #5

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: the polar of an external point · two equal tangents · $\sin \frac{\theta}{2} = \frac{r}{OP}$ · the director circle.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

Read OP and r straight off, then the tangent length is one step of Pythagoras.

- Write down the length of the tangent from $(0, 5)$ to $x^2 + y^2 = 9$.
- The point $(0, 5)$ sits outside $x^2 + y^2 = 9$ with $OP = 5$ and $r = 3$. Write down $\tan \frac{\theta}{2}$, where θ is the angle between the two tangents.
- Write down the equation of the chord of contact (polar) drawn from the point $(5, 3)$ to $x^2 + y^2 = 25$.
- The circle $x^2 + y^2 = 25$ has director circle $x^2 + y^2 = 2r^2$. Write down the radius of that director circle.
- Write down the length of the tangent from $(7, 0)$ to $x^2 + y^2 = 36$.
- Two tangents from the point P to a circle of unknown radius make an angle of 60° and $OP = 6$. Write down the length of each tangent.
- Write down the equation of the chord of contact drawn from $(5, 0)$ to $x^2 + y^2 = 4$ (the two tangency points join in a vertical line).
- A tangent from the point P to $x^2 + y^2 = 25$ has length 12. Write down OP .
- The two tangents from P to the circle $x^2 + y^2 = 25$ make an angle of 120° . Write down OP .
- The chord of contact from the point P to $x^2 + y^2 = 16$ is the line $x + y = 4$. Write down P .

Section B Manipulation Drills

(≤50 s each)

Feed external points and radii through $\sqrt{OP^2 - r^2}$ and the polar formula, choosing from the options.

- The length of the tangent from $(3, 4)$ to $x^2 + y^2 = 9$ is?

A) $\sqrt{7}$

- B) 4
C) 5
D) 2
2. The point P has $OP = 6$ with respect to a circle of radius 3. The angle between the two tangents from P is?
A) 30°
B) 45°
C) 60°
D) 90°
3. Write down the equation of the chord of contact drawn from the point $(3, 5)$ to $x^2 + y^2 = 25$.
4. The director circle of $x^2 + y^2 = 4$ has radius?
A) 2
B) $2\sqrt{2}$
C) 4
D) $\sqrt{2}$
5. The length of the tangent from $(1, 2)$ to $x^2 + y^2 = 1$ is?
A) 1
B) $\sqrt{5}$
C) 2
D) $\sqrt{3}$
6. A tangent from P to a circle of radius 5 has length 12. How far is P from the centre?
A) 17
B) 7
C) 13
D) $\sqrt{119}$
7. The chord of contact from $P = (a, 0)$ to $x^2 + y^2 = 16$ is the line $x = 4$. Find a .
A) 2
B) 4
C) 8
D) 16
8. The two tangents from P to $x^2 + y^2 = 25$ meet at right angles (so P lies on the director circle). Write down the length of each tangent.

9. The tangent to $x^2 + y^2 = 8$ with slope 1 has positive y -intercept equal to?
- A) 4
 - B) $2\sqrt{2}$
 - C) 8
 - D) $4\sqrt{2}$
10. A tangent from P to a circle has length 6, and $OP = 10$. Write down the circle's radius.

Section C Substitution & Structure(≤ 75 s each)

Invert the tools: reconstruct OP from a tangent length, or the polar from a chord, reading circle structure first.

1. A tangent from P to $x^2 + y^2 = 9$ has length $\sqrt{7}$. Find OP .
- A) 4
 - B) 5
 - C) $\sqrt{34}$
 - D) 8
2. From points P the tangent to $x^2 + y^2 = 4$ equals the tangent to $(x - 6)^2 + y^2 = 16$ in length. The locus of P is their radical axis. Find its equation.
- A) $x = 2$
 - B) $x = 3$
 - C) $x = 4$
 - D) $x = 6$
3. The chord of contact from $P = (3, 4)$ to $x^2 + y^2 = 4$ has equation $3x + 4y = 4$. What is its distance from the origin?
- A) $\frac{5}{4}$
 - B) $\frac{4}{5}$
 - C) 1
 - D) $\frac{3}{5}$
4. The circle $x^2 + y^2 - 6x - 8y + 24 = 0$ has centre $(3, 4)$ and radius 1. What is the length of the tangent from $(6, 8)$?
- A) $2\sqrt{6}$
 - B) $2\sqrt{5}$
 - C) 4
 - D) $\sqrt{26}$

5. The circle $(x - 4)^2 + y^2 = 9$ has director circle of radius?
- A) $3\sqrt{2}$
 - B) 6
 - C) 18
 - D) 9
6. The length of the tangent from $(8, 6)$ to $x^2 + y^2 = 1$ is?
- A) $3\sqrt{11}$
 - B) $\sqrt{99}$
 - C) 10
 - D) $9\sqrt{11}$
7. For a 13-14-15 triangle the incircle touches each side, and the two tangent segments from a vertex are equal to $s - a$ (semiperimeter minus the opposite side). The vertex opposite the side of length 13 has tangent segments of length?
- A) 4
 - B) 8
 - C) 12
 - D) 21
8. The chord of contact (polar) of the point $(2, 1)$ with respect to the unit circle $x^2 + y^2 = 1$ is the line $2x + y = 1$. What is its distance from the origin?
- A) $\frac{\sqrt{5}}{5}$
 - B) $\frac{1}{3}$
 - C) 1
 - D) $\frac{\sqrt{5}}{2}$

Section D Challenge Ramp(≤ 90 s each)

Stack the tangent length with the angle rule and the polar, folding in the half-chord and incircle ideas from earlier days.

1. Two tangents are drawn to $x^2 + y^2 = 9$ from a point P with $OP = 6$. What is the angle between them?
- A) 30°
 - B) 45°
 - C) 60°
 - D) 90°

2. A tangent from P to $x^2 + y^2 = 25$ has length $5\sqrt{3}$. How far is P from the centre?
- A) 10
B) $5\sqrt{3}$
C) $5\sqrt{6}$
D) $\frac{20}{\sqrt{3}}$
3. The chord of contact from $P = (3, 3)$ to $x^2 + y^2 = 9$ has equation $x + y = 3$. What is the length of this chord?
- A) $3\sqrt{2}$
B) 3
C) $\sqrt{18}$
D) $6\sqrt{2}$
4. The two tangents from P to a circle make an angle of 60° and $OP = 10$. What is the length of each tangent?
- A) 10
B) $5\sqrt{3}$
C) 5
D) $\frac{10}{\sqrt{3}}$
5. The chord of contact (polar) of the point $(9, 12)$ with respect to $x^2 + y^2 = 81$ is the line $9x + 12y = 81$. What is its distance from the origin?
- A) $\frac{27}{5}$
B) 5
C) $\frac{9}{5}$
D) $\frac{27}{13}$

“A tangent is an archer’s draw: Pythagorean strength $OP^2 = r^2 + (\text{tangent})^2$ fired down a chord of contact that is just the same formula straightened.”

Found a mistake or want to contribute a sheet?
Visit the open-source project at github.com/The-CerealDev/speed-maths